Name- Chaitakshi
User Id- 1002101144
Class - X^{th}
Subject - Maths

1. Monomial

2. Let $P(x) = (x+1)(x^2 - x - x^4 + 1)$

$x^3 - \cancel{x^2} - x^5 + \cancel{x} + \cancel{x^2} - \cancel{x} - x^4 + 1$

$= x^3 - x^5 - x^4 + 1$

So, the degree of the polynomial is 5.

3. $f(x) = a^2x^2 - 3ax + 3x - 1$

zero of the polynomial $= 1$

$f(1) = a^2 1 - 3a + 3 - 1 = 0$

$= a^2 - 3a + 2 = 0$

$= a^2 - 2a - a + 2 = 0$

$= a(a-2) - 1(a-2) = 0$

$(a-2)(a-1) = 0$

$a - 2 = 0 \quad , a = 2$

$a - 1 = 0 \quad , a = 1$

So, the value of a is 1, 2.

4. The number of zero in the following graph is 1

5. let α be the one root of quadratic equation
So, other root $= \frac{1}{\alpha}$

$f(x) = 2x^2 - 3x + k$
but multiplication of two roots $= \frac{c}{a}$

$\alpha \times \frac{1}{\alpha} = \frac{k}{2}$

$k = 2$ Ans

6. Sum of zeroes $= 5$
Product of zeroes $= -14$

$p(x) = x^2 - (\alpha + \beta)x + (\alpha\beta)$
$= x^2 - (5)x + (-14)$
$= x^2 - 5x - 14$ Ans.

7. $p(x) = x^2 + 5x + 6$
$= \cancel{x^2 + 3x + 2x}$ $x^2 + 3x + 2x + 6$
$= x(x+3) + 2(x+3)$
$= (x+2)(x+3)$

$x + 2 = 0$, $x + 3 = 0$
$x = -2$ $x = -3$

Verification
Sum of zeroes $= (-2) + (-3) = -5$
$\frac{-b}{a} = \frac{-5}{1} = -5$

Product of zeroes $= (-2) \times (-3) = 6$

$$\frac{c}{a} = \frac{6}{1} = 6.$$

verified.

8. Zeroes of polynomial $= 3, 5$ and -2

$$p(x) = x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - (\alpha\beta\gamma) = 0$$
$$= x^3 - (3 + 5 + (-2))x^2 + (3\times5 + 5\times(-2) + (-2)\times3)x$$
$$- (3\times5\times(-2)) = 0$$
$$= x^3 + 6x^2 + (15 - 10 - 6)x - (-30) = 0$$
$$= x^3 + 6x^2 - 1x + 30 = 0.$$ Ans.

9. $x + \frac{1}{x} = 3$

To find $= x^3 + \frac{1}{x^3}$

Cubing on both sides,

$$\left(x + \frac{1}{x}\right)^3 = (3)^3$$

$$= x^3 + \frac{1}{x^3} + 3 \times \cancel{x} \times \frac{1}{\cancel{x}}\left(x + \frac{1}{x}\right) = 27$$

$$= x^3 + \frac{1}{x^3} + 3 \times 3 = 27$$

$$= x^3 + \frac{1}{x^3} = 27 - 9$$

$$x^3 + \frac{1}{x^3} = 18.$$

10. If α and β are the zeroes of the quadratic polynomial $f(x) = 15x^2 - 5x + 6$.

$(\alpha+\beta) = \frac{-b}{a} = \frac{-(-5)}{15} = \frac{1}{3}$

$\alpha\beta = \frac{c}{a} = \frac{\cancel{6}^2}{\cancel{15}_5} = \frac{2}{5}$

Now,

$\left(1+\frac{1}{\alpha}\right)\left(1+\frac{1}{\beta}\right) = \left(\frac{\alpha+1}{\alpha}\right)\left(\frac{\beta+1}{\beta}\right)$

$= \frac{(\alpha+1)(\beta+1)}{\alpha\beta} = \frac{\alpha\beta+\alpha+\beta+1}{\alpha\beta}$

$= \frac{\frac{2}{5}+\frac{1}{3}+1}{\frac{2}{5}}$

$= \frac{\frac{6+5+15}{15}}{\frac{2}{5}}$

$= \frac{\cancel{26}^{13}}{\cancel{15}_3} \times \frac{\cancel{5}^1}{\cancel{2}_1} = \frac{13}{3}$ Ans